

ICHTHYOSIS VULGARIS

Ichthyosis vulgaris (OMIM #146700), the most common form of ichthyosis, is typically mild. Although skin appears normal at birth, the condition usually becomes apparent within the first year of life. Scaling is most prominent on the extensor surfaces of the limbs, with relative sparing of flexural areas and the diaper region. The scales are often fine and white, covering large areas of the body.

On the lower extremities—frequently the most affected area—the scales may be centrally adherent with elevated edges, creating superficial fissures (cracking) in the stratum corneum. This gives the skin a rough texture.

Associated Findings

Several additional features are commonly seen in ichthyosis vulgaris:

- **Hyperlinear palms:** Increased skin markings on the palms are typically present.
- **Palmar/plantar hyperkeratosis:** Some individuals exhibit thickening that may approach keratoderma.
- **Keratosis pilaris:** A follicular hyperkeratosis presenting as spine-like projections, commonly affecting the lateral arms, extensor thighs, and buttocks.
- **Atopy:** There is a strong association with atopic conditions such as eczema, asthma, and hay fever.
- **Hypohidrosis:** Rarely, reduced sweating may occur, leading to heat intolerance.

It is important to note that hyperlinear palms and keratosis pilaris can also be seen in individuals with atopy without ichthyosis vulgaris, which may complicate diagnosis.

Clinical Variability and Environmental Influence

There is considerable variability in disease severity, even among affected members of the same family. Symptoms typically worsen in cold, dry climates and improve significantly in warm, humid environments—sometimes to the point of near-complete clearance.

Genetics

Ichthyosis vulgaris is caused by loss-of-function mutations in the *FLG* gene, which encodes profilaggrin. Inheritance follows a **semi-dominant** pattern:

- **Homozygotes or compound heterozygotes** (mutations in both *FLG* alleles): exhibit a **severe phenotype**.
- **Heterozygotes** (mutation in one allele): present with a **milder phenotype**, with approximately **60% penetrance**.

The carrier frequency for *FLG* mutations is about **9%**, and the prevalence of homozygotes is estimated at **1 in 400** in the Anglo-European population.

Diagnostic Challenges

- Mild ichthyosis vulgaris can be difficult to distinguish from simple **xerosis** (dry skin).
- Severe cases in males may resemble **X-linked recessive ichthyosis**, making clinical differentiation challenging without genetic testing.
- Histopathology (Fig. 47-2A) may show characteristic findings, but overlap with other conditions can occur.

Ongoing research continues to clarify the spectrum of *FLG*-related skin barrier dysfunction, ranging from mild dry skin to full-blown ichthyosis vulgaris.